EXPERIMENT NO: 16

Experiment to understand EDA - Quantitative and Qualitative analysis

Program:

```
import seaborn as sns
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline
tips=sns.load_dataset('tips')
print(tips.head())
sns.displot(tips.total_bill,kde=True)
sns.displot(tips.total_bill,kde=False)
sns.jointplot(x=tips.tip,y=tips.total_bill)
sns.jointplot(x=tips.tip,y=tips.total_bill,kind="reg")
sns.jointplot(x=tips.tip,y=tips.total_bill,kind="hex")
sns.pairplot(tips)
print(tips.time.value_counts())
sns.pairplot(tips,hue='time')
sns.pairplot(tips, hue='day')
plt.show()
plt.close()
sns.heatmap(tips.corr(numeric_only=True), annot=True)
plt.show()
plt.close()plt.figure()
sns.boxplot(y=tips.total_bill)
plt.show()
plt.close()
plt.figure()
sns.boxplot(y=tips.tip)
plt.show()
plt.close()

plt.figure()
sns.countplot(x=tips.day)
plt.show()
plt.close()
```

```
plt.figure()
sns.countplot(x=tips.sex)
plt.show()
plt.close()

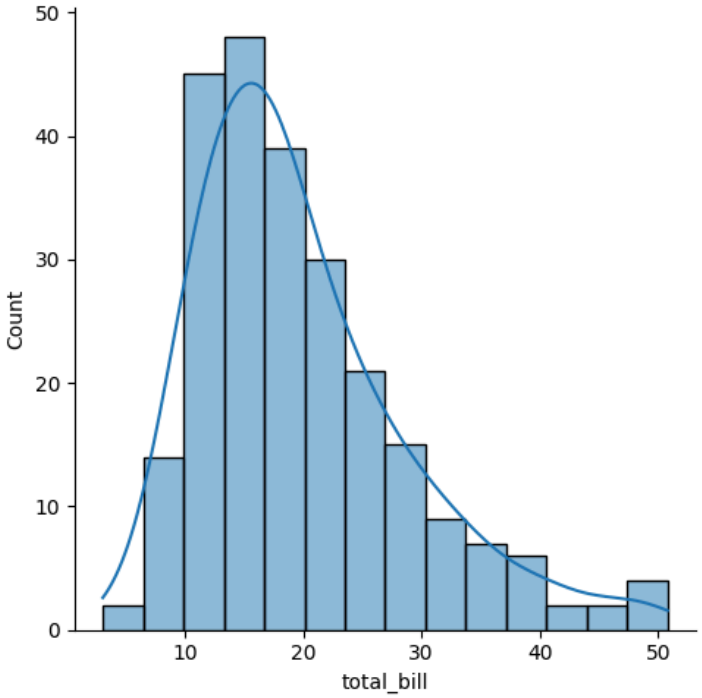
plt.figure()
tips.sex.value_counts().plot(kind='pie')
plt.show()
plt.close()

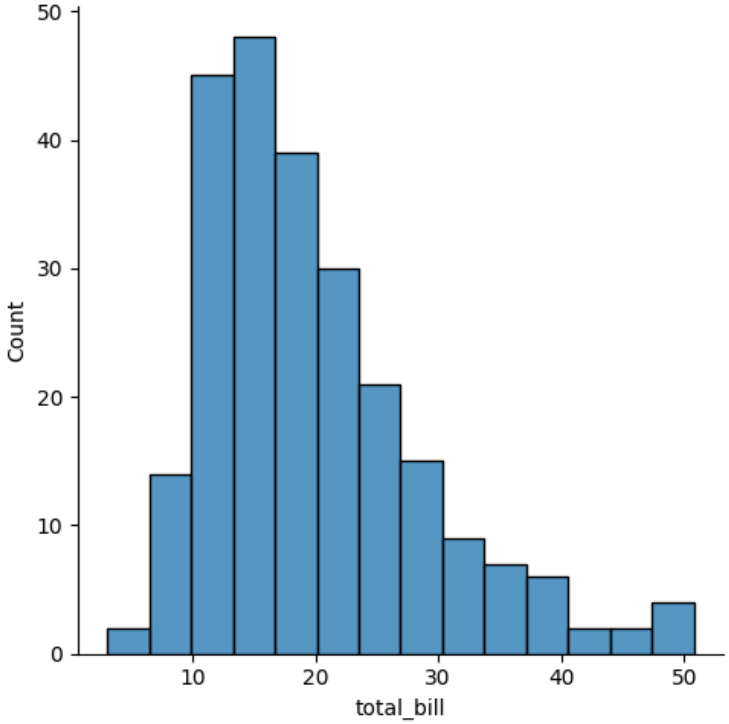
plt.figure()
tips.sex.value_counts().plot(kind='bar')
plt.show()
plt.close()

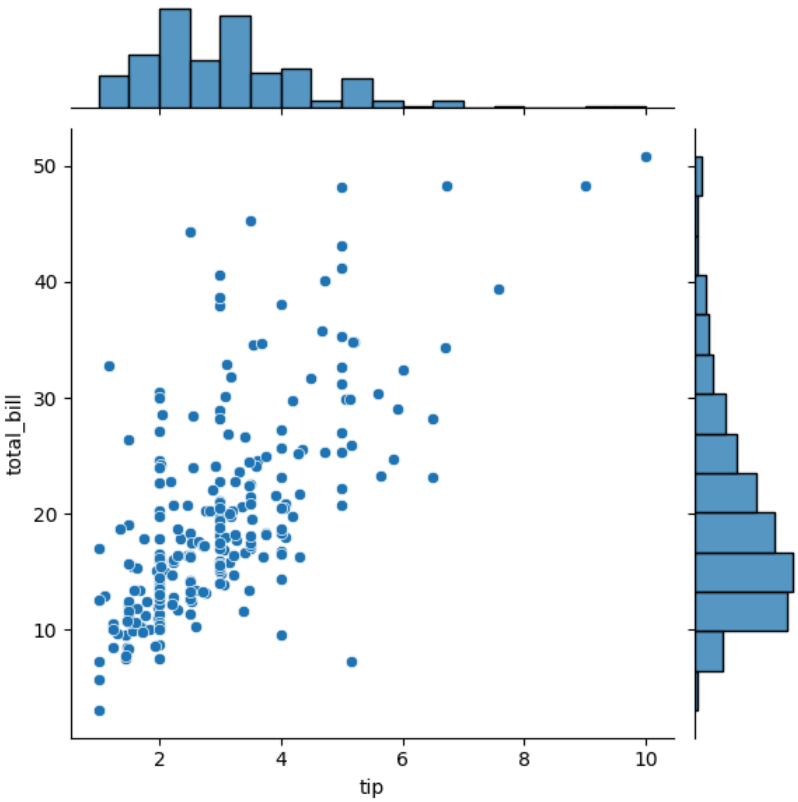
plt.figure()
sns.countplot(x=tips[tips.time == 'Dinner']['day'])
plt.show()
plt.close()
```

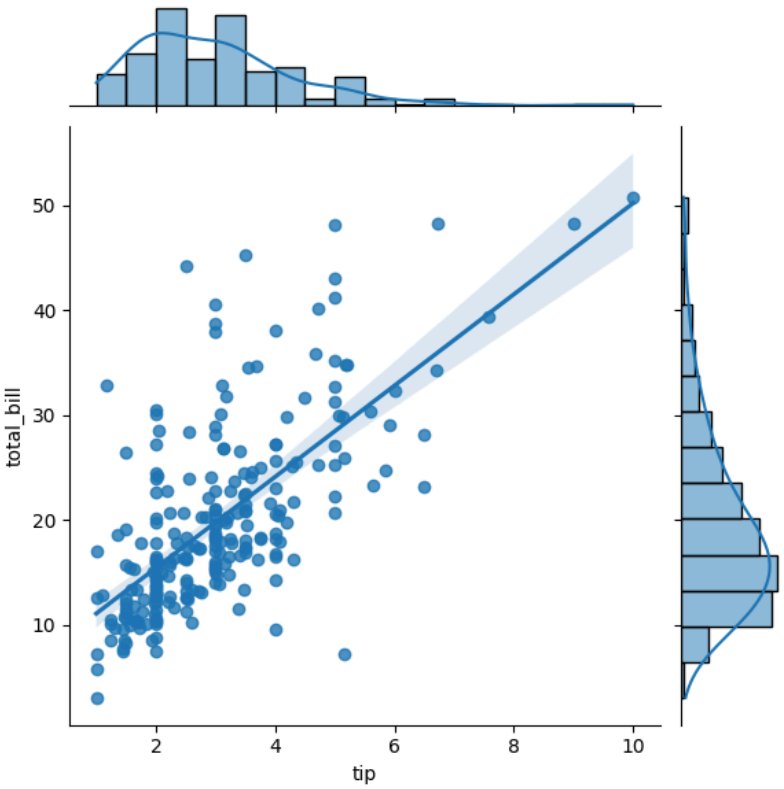
Output:

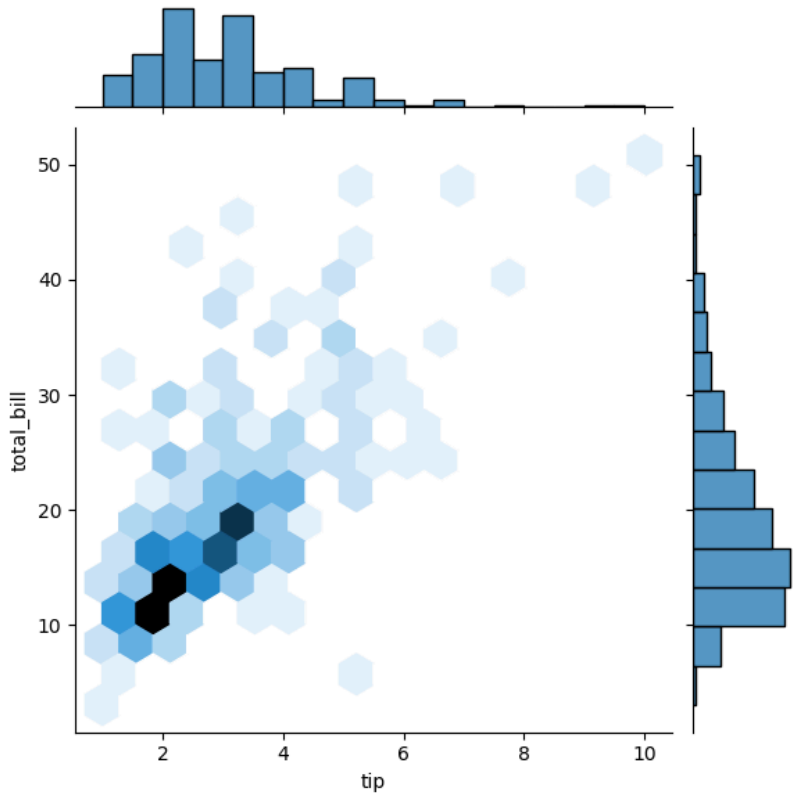
```
  total_bill   tip     sex smoker  day    time  size
0      16.99  1.01  Female     No  Sun  Dinner     2
1      10.34  1.66    Male     No  Sun  Dinner     3
2      21.01  3.50    Male     No  Sun  Dinner     3
3      23.68  3.31    Male     No  Sun  Dinner     2
4      24.59  3.61  Female     No  Sun  Dinner     4
time
Dinner    176
Lunch      68
Name: count, dtype: int64
```

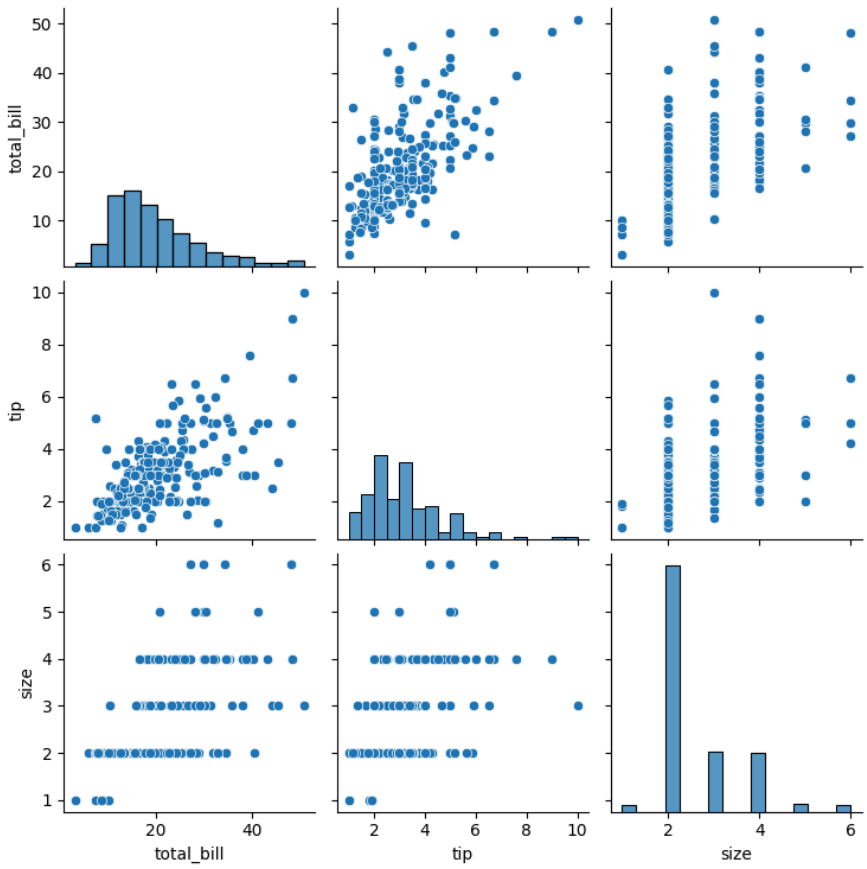

Count
total_bill
0
10
20
30
40
50

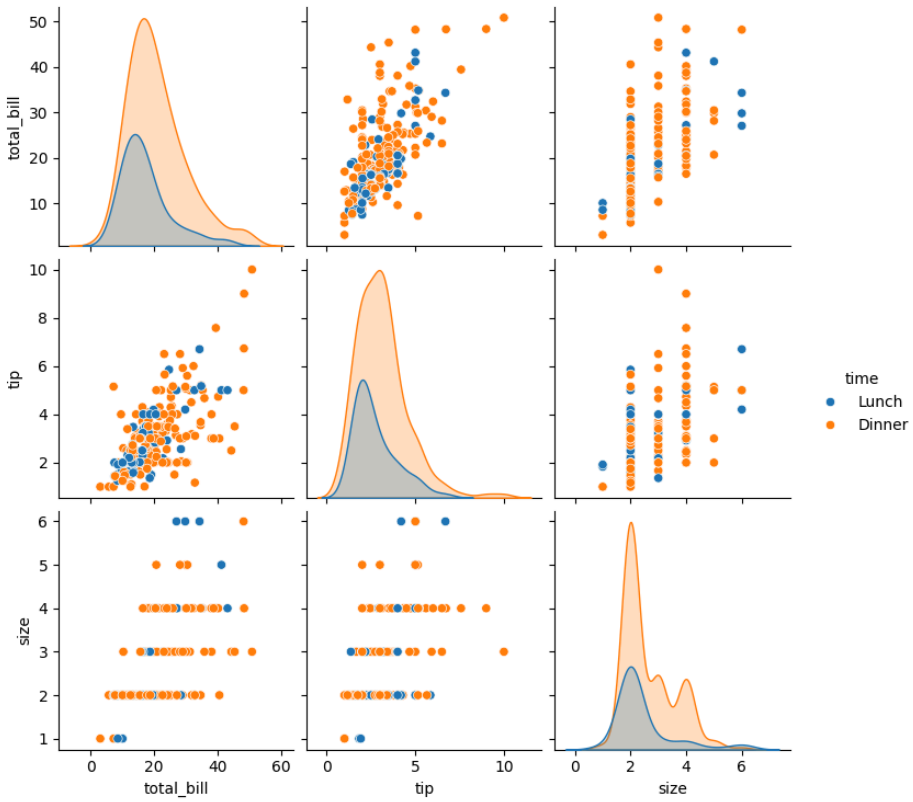

Count
total_bill
0
10
20
30
40
50

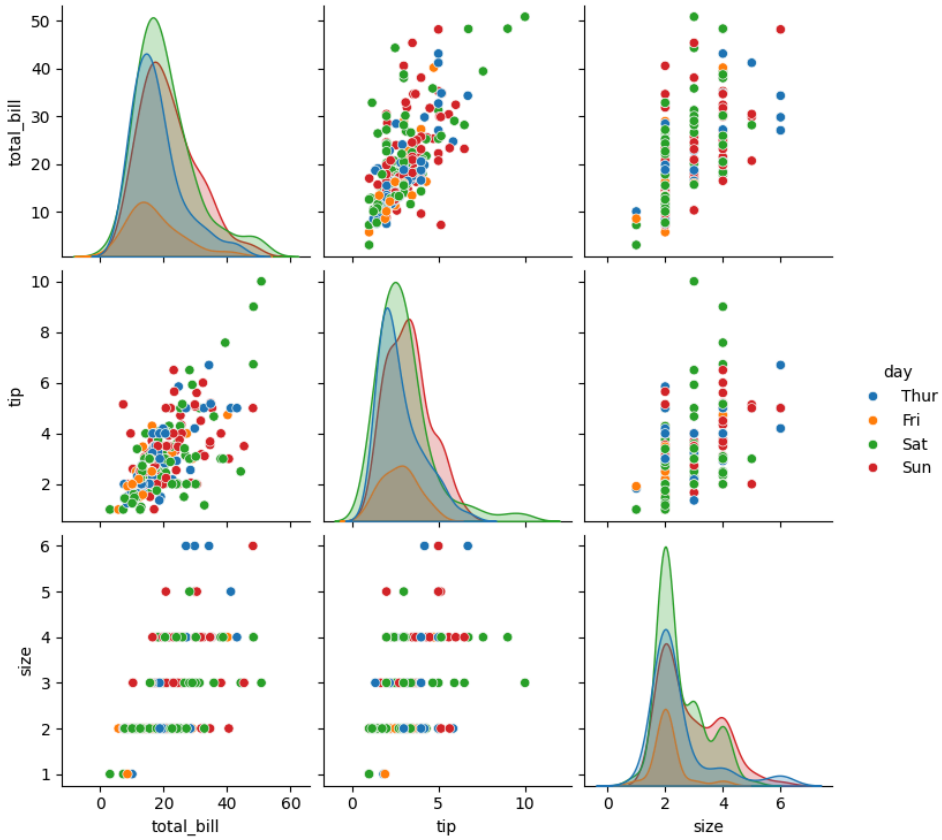

total_bill
tip
50
40
30
20
10
2
4
6
8
10

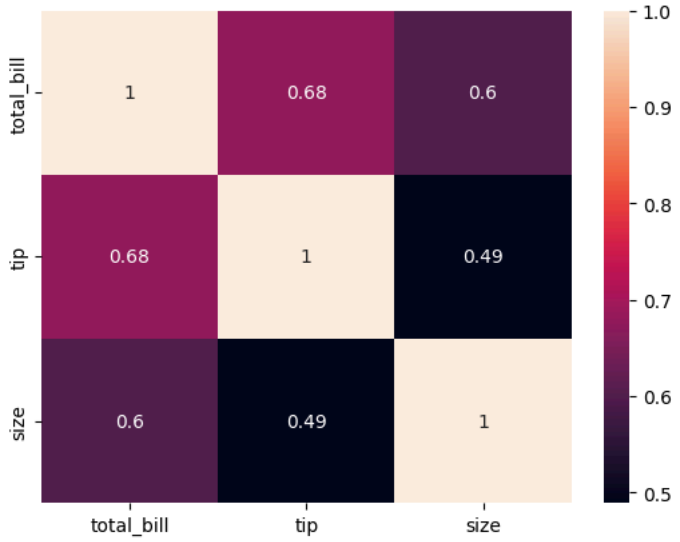

total_bill
tip
50
40
30
20
10
2
4
6
8
10

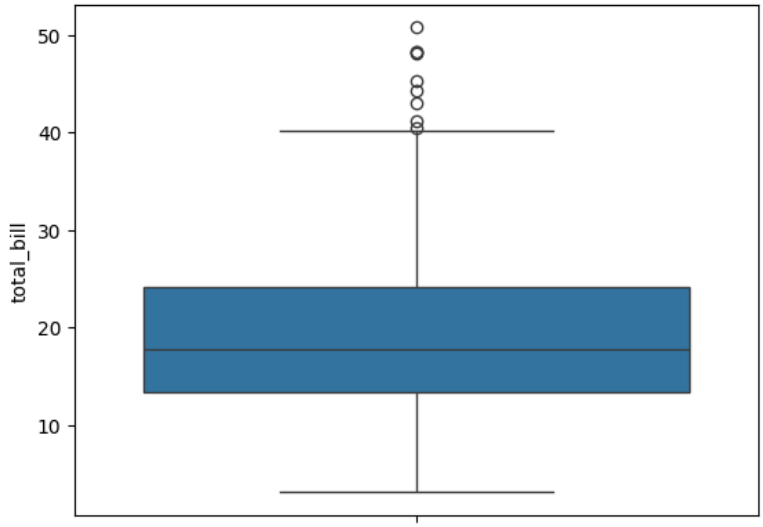

total_bill
tip
50
40
30
20
10
2
4
6
8
10

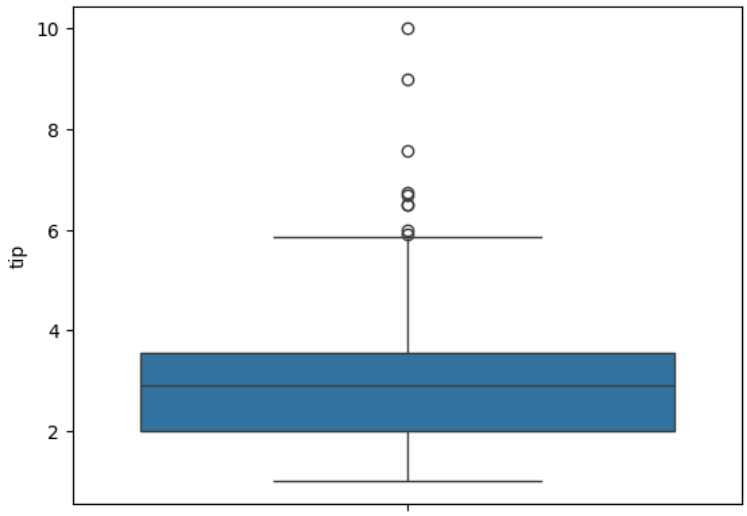

total_bill
tip
size
50
40
30
20
10
10
8
6
4
2
6
5
4
3
2
1
20
40
2
4
6
8
10
2
4
6

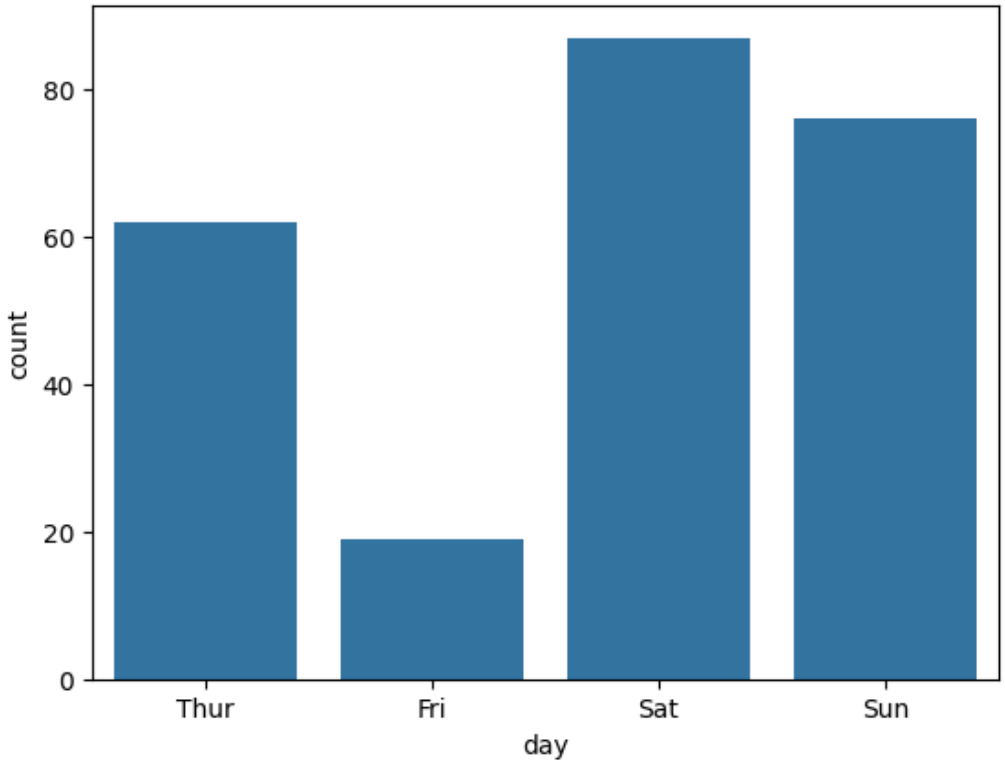

total_bill
tip
size
time
Lunch
Dinner

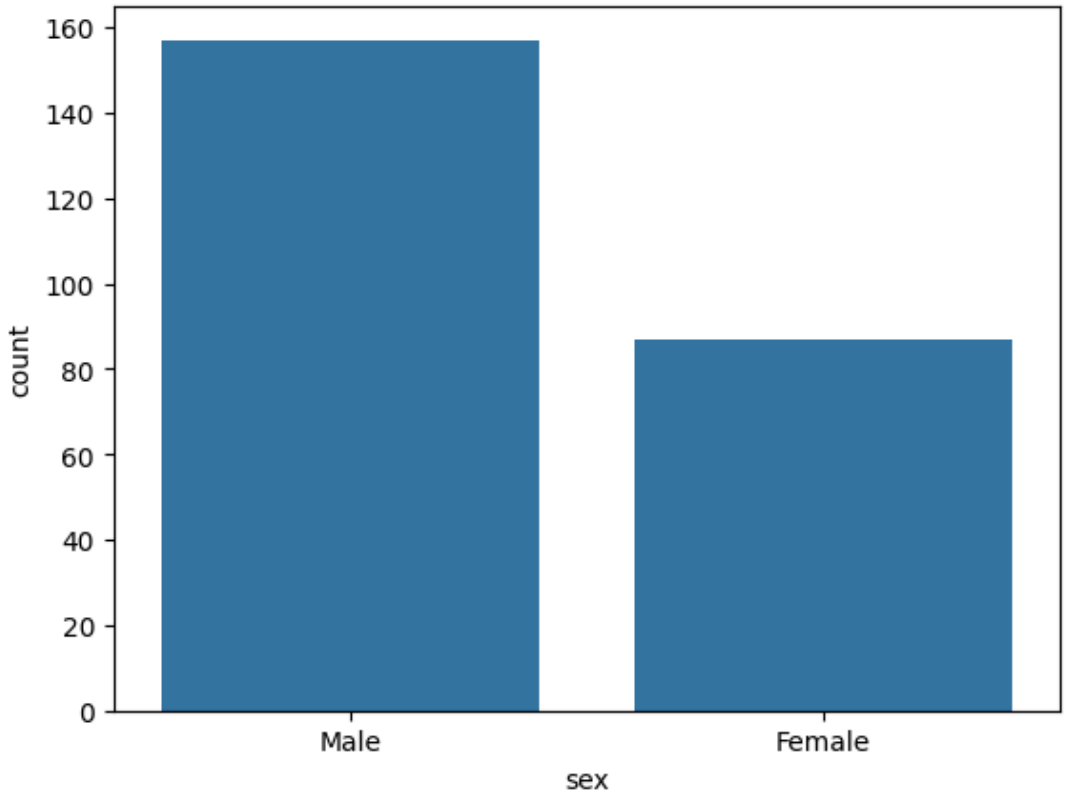

total_bill
tip
size
day
Thur
Fri
Sat
Sun

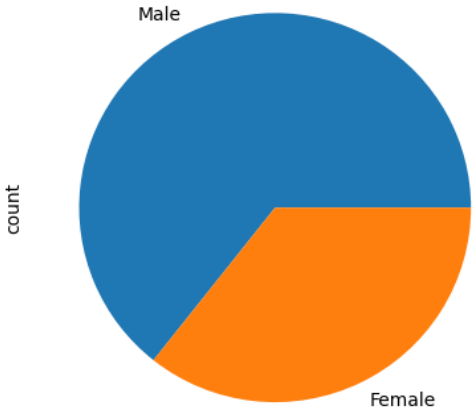

total_bill
tip
size
1
0.68
0.6
0.68
1
0.49
0.6
0.49
1
total_bill
tip
size
1.0
0.9
0.8
0.7
0.6
0.5

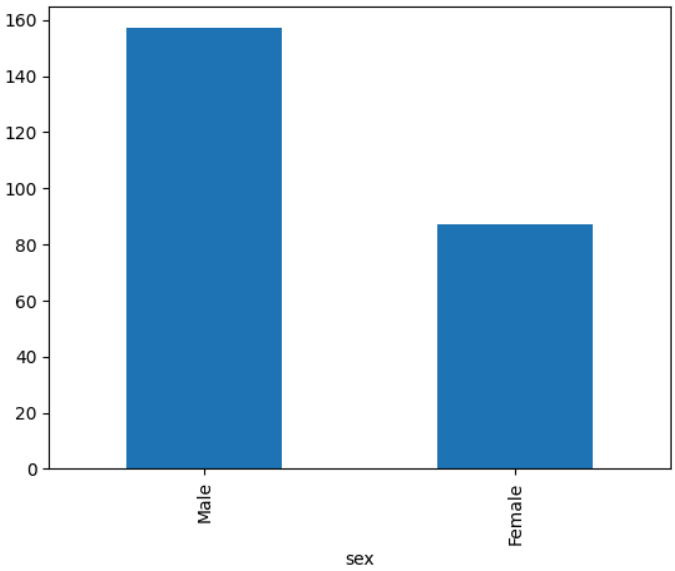

total_bill
50
40
30
20
10

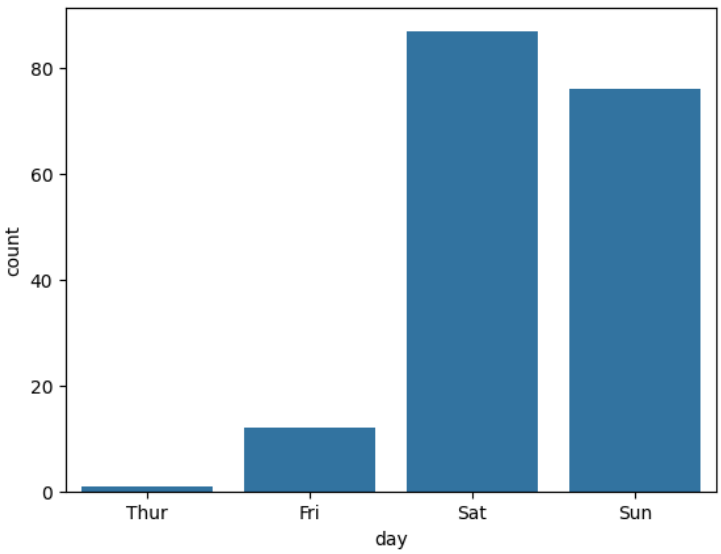

tip
10
8
6
4
2


80
60
40
20
0
count
Thur
Fri
Sat
Sun
day


160
140
120
100
80
60
40
20
0
count
Male
Female
sex


Male
count
Female


160
140
120
100
80
60
40
20
0
Male
Female
sex


80
60
40
20
0
count
Thur
Fri
Sat
Sun
day